Review of *Multigroup First Crack* (26 January 2026)

Equation numbers refer to the original document. Items are ordered by importance within each section.

Variable Basis Method (Section 3)

1. Eq. (42) is incorrect (main error)

The text says T cancels from both sides, but the left and right sides were scaled by different factors. Multiplying (41) by T gives

$$\frac{1}{c}\left(\frac{\partial\hat{\psi}}{\partial t}-\frac{u}{T}\frac{\partial T}{\partial t}\frac{\partial\hat{\psi}}{\partial u}-\frac{\hat{\psi}}{T}\frac{\partial T}{\partial t}\right)+\Omega_x\left(\frac{\partial\hat{\psi}}{\partial x}-\frac{u}{T}\frac{\partial T}{\partial x}\frac{\partial\hat{\psi}}{\partial u}-\frac{\hat{\psi}}{T}\frac{\partial T}{\partial x}\right)=4\pi\sigma_a\hat{B}-\sigma_a\hat{\psi}.$$

The left side of (42) as written corresponds to multiplying (41) by T^2, but the right side was only multiplied by T. A units check confirms the mismatch: the two sides of (42) differ by a factor of T. If the left side of (42) is kept as written, the right side must be

$$T\left(4\pi\sigma_a\hat{B}-\sigma_a\hat{\psi}\right).$$

This error carries into Eqs. (43) and (45), whose right sides should be $cT\sigma_a\left(4\pi\hat{B}-\hat{\psi}\right)$.

2. Recommended conservative form

Using $u\,\partial_u\hat{\psi}+\hat{\psi}=\partial_u(u\hat{\psi})$, the two $\hat{V}$ terms combine. After dividing by T:

$$\frac{1}{c}\frac{\partial\hat{\psi}}{\partial t}+\Omega_x\frac{\partial\hat{\psi}}{\partial x}-\frac{\hat{V}}{cT}\frac{\partial(u\hat{\psi})}{\partial u}=\sigma_a\left(4\pi\hat{B}-\hat{\psi}\right),\qquad\hat{V}=\frac{\partial T}{\partial t}+c\,\Omega_x\frac{\partial T}{\partial x}.$$

This can be checked directly. Let $L=\frac{1}{c}\partial_t+\Omega_x\partial_x$ at fixed ν. Then

$$L\left(\frac{\hat{\psi}}{T}\right)=\frac{1}{T}\left[L\hat{\psi}\big|_u-\frac{LT}{T}\frac{\partial(u\hat{\psi})}{\partial u}\right],$$

with $LT=\hat{V}/c$.

The divergence form in u is the right starting point for a group discretization in u, because it conserves energy across groups. Some consequences:

- $\hat{V}$ depends on angle through Ω_x.
- The sign of $\hat{V}$ sets the upwind direction in u.
- $u\hat{\psi}\to 0$ at $u=0$, so no boundary condition is needed there.

3. $\hat{\psi}$ must depend on x and Ω

Eqs. (32)–(45) write $\hat{\psi}(u,t)$. If that were literally true, $\partial\hat{\psi}/\partial x$ would vanish. It should be $\hat{\psi}(u,x,t,\Omega)$.

4. Notation in Eqs. (37)–(38)

As written, $\partial_t\hat{\psi} = \partial_t\hat{\psi} + \partial_u\hat{\psi}\,\partial_t u$ implies the last term is zero. The two derivatives must be distinguished: the left side is taken at fixed ν, the right side at fixed u. For example,

$$\left.\frac{\partial\hat{\psi}}{\partial t}\right|_\nu = \left.\frac{\partial\hat{\psi}}{\partial t}\right|_u + \left.\frac{\partial\hat{\psi}}{\partial u}\,\frac{\partial u}{\partial t}\right|_\nu, \qquad \left.\frac{\partial u}{\partial t}\right|_\nu = -\frac{u}{T}\frac{\partial T}{\partial t}.$$

With this interpretation, Eqs. (35)–(41) are correct.

5. Eq. (33) uses the wrong symbol

It should read $\psi(\nu,x,t) = \hat{\psi}(u,x,t)/T(x,t)$. This is the angular intensity ψ, not the scalar flux ϕ.

6. Define $\hat{B}$ explicitly

From $\hat{B}\,du = B\,d\nu$:

$$\hat{B}(u,T) = T\,B(uT,T) = T^4\,\frac{2hu^3/c^2}{e^{hu/k}-1}.$$

This shows the motivation for the method: in u the Planck spectrum has a fixed shape that only scales as T^4. Note also that the opacity becomes $\sigma_a(uT,T)$.

7. Eq. (46) mixes variables

It integrates over ν but uses $\hat{\phi}(u)$ and the group quantity $\bar{B}_g$. Using $\phi\,d\nu = \hat{\phi}\,du$, it should read

$$C_v\frac{\partial T}{\partial t} = \int_0^\infty du\,\sigma_a(uT,T)\left[\hat{\phi}(u,x,t) - 4\pi\hat{B}(u,T)\right], \qquad \hat{\phi} = \int\hat{\psi}\,d\Omega.$$

Also, $T(t)$ should be $T(x,t)$.

8. Practical caveat

$u = \nu/T$ is singular at $T = 0$, and $\hat{V}$ contains $\partial_x T$, which is very large at a Marshak wave front. A cold initial condition will need a temperature floor, and the front will need care in the discretization.

Gray and Multigroup Formulation (Sections 1–2)

9. The 4π factors are inconsistent

Eq. (1) has $4\pi\sigma_a B$, while Eq. (3) has $\sigma_a B$. The material equation (2), with $\phi = \int\psi\,d\Omega$, is consistent with B being the Planck *intensity*. In that case the transport emission source should be $\sigma_a B$, with no 4π. One convention should be chosen and applied to Eqs. (1), (3), (5), (14), (18), (25), (28), (34), and (45).

10. Eq. (15) has an extra 4π

Since $\int_0^\infty B\,d\nu = acT^4/(4\pi)$, the emission term is

$$4\pi \int_0^\infty \sigma_a B\,d\nu = \sigma_P\, acT^4,$$

not $4\pi\sigma_P\, acT^4$. The same correction applies to Eq. (16).

11. Eq. (10) uses the wrong cross section

It has σ_a where it should have σ^*. Eq. (11) correctly uses σ^*.

12. Sign error for $\mu_l < 0$ in Eqs. (12) and (30)

Starting from the upwind difference (9):

$$\mu_l \frac{\psi_{i+1} - \psi_i}{\Delta x} + \sigma^* \psi_i = Q^* \implies \psi_i = \frac{Q^* - \mu_l \psi_{i+1}/\Delta x}{\sigma^* - \mu_l/\Delta x} = \frac{Q^* + |\mu_l|\,\psi_{i+1}/\Delta x}{\sigma^* + |\mu_l|/\Delta x}.$$

As written, the denominator $\mu_l/\Delta x + \sigma^*$ can be zero or negative.

13. Eq. (22) is not an average

$\int \sigma_a\,d\nu$ has units of opacity $\times$ frequency. For Eq. (24) and the emission term to make sense, $\bar{\sigma}_{ag}$ must be a Planck-weighted average:

$$\bar{\sigma}_{ag} = \frac{\int_{\nu_{g-1/2}}^{\nu_{g+1/2}} \sigma_a B\,d\nu}{\int_{\nu_{g-1/2}}^{\nu_{g+1/2}} B\,d\nu}.$$

With this definition, $\int_g \sigma_a B\,d\nu = \bar{\sigma}_{ag}\bar{B}_g$ holds exactly, and Eq. (24) is an approximation. Also, the group bounds are written $\nu_{i\pm1/2}$, which clashes with the spatial index i; they should be $\nu_{g\pm1/2}$.

14. Eq. (28) is incorrect

It includes $-\bar{\sigma}_{ag}\bar{\psi}_g$, which is already in σ^*, so absorption is counted twice. It also drops the old-time-level term. It should be

$$Q^*_{i,g,l} = 4\pi\bar{\sigma}_{ag}\bar{B}_g(T_i^n) + \frac{\bar{\psi}_{i,g}^{n,l}}{c\Delta t}$$

(with the 4π convention fixed per item 9).

15. Eq. (31) treats emission explicitly

Eq. (31) evaluates emission at $B(T^n)$. This conserves energy, because the radiation emits the same amount the material loses. However, it can oscillate or become unstable when Δt is large compared with $C_v/(4\sigma_P acT^3)$. The linearization in Eq. (17) is introduced but never used to close the system, for example through a Fleck factor or effective-scattering formulation.

Minor Notation

- Eqs. (8)–(9): the subscript on μ should be l, not i.
- Eq. (13): stray comma in $\psi_{i+1/2,}$.
- Eq. (14): the superscript l on σ_a and B denotes angle, but these are angle-independent; probably the group index was intended.
- Eq. (17): mixed sub/superscripts on n; should be $(T^4)^{n+1} \approx (T^n)^4 + 4(T^n)^3(T^{n+1} - T^n)$.
- Eq. (20): the "=" sits inside the integral's parentheses.
- Eq. (27): $\bar{\phi}(\nu, t, x)$ should be $\bar{\phi}_g(x, t)$.
- Eqs. (2), (19), (46): $T(t)$ should be $T(x, t)$.
- Eq. (7): calling it the "steady state eq" is misleading; it is the problem solved within each time step.

Figure 1 (Reed's Problem)

Reed's problem includes scattering regions, but none of the equations in the document have a scattering term. The text also says the test problems used a constant Q^*. The figure therefore tests only the transport sweep, not the material coupling. The document should explain how scattering was handled.